



Thermal Properties 1

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Week 12 lectures

Thermal properties

- Heat capacity and specific heat
- Heat conduction and thermal conductivity
- Thermo-mechanical behaviour
- Thermal expansion
- Thermal stress and thermal shock

Background information:

For these lectures read

Callister and Rethwisch 9e Chapter 20

Aims

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Learning outcomes



Briefly explain the phenomenon of thermal expansion from an atomic perspective using a potential energy-versus-interatomic separation



Define heat capacity and specific heat capacity.



Define thermal conductivity



Outline the principal mechanisms of thermal conductivity and compare those of metals, ceramics, and polymers



Note the primary mechanism by which thermal energy is assimilated in solid materials.



Explain the linear coefficient of thermal expansion.

Thermal behaviour

- **Thermal behavior** could be defined as the response of a material to thermal energy or changing temperatures (heat)

2 temperatures are obviously important to materials:

- 1) Melting temperature
- 2) Glass transition temperature / ductile-brittle transition

These transitions may be well defined:

e.g. sharp melting temperature of an alloy at the eutectic point

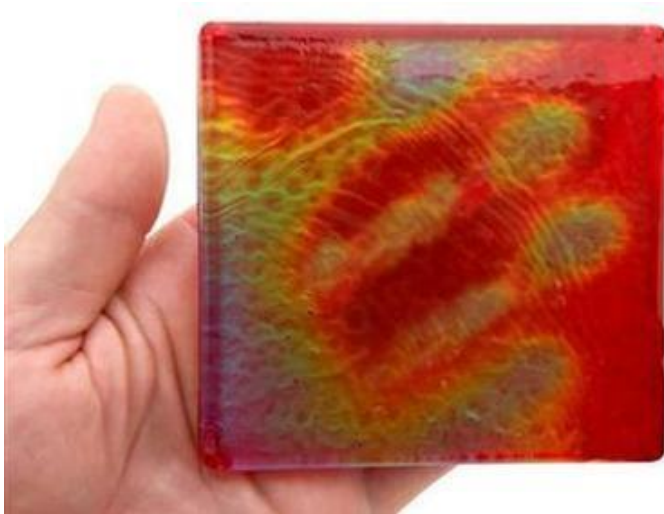
Or may transition over a range of temperature:

e.g. glass transition in semi-crystalline polymers

Thermal behaviour

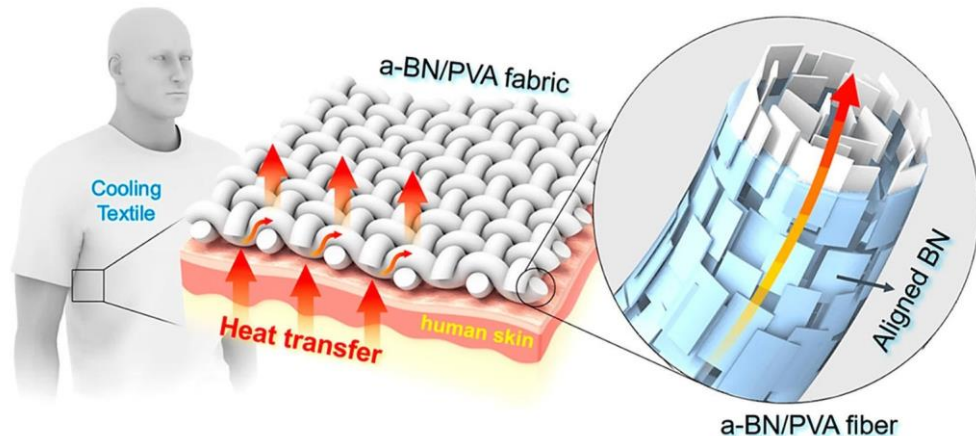
Materials for thermal applications

- Thermal insulation
 - Heat transfer
 - Thermal responsive materials
- } Thermal management



Thermo-chromic materials
for temperature sensing

Thermal regulation textile with highly aligned nanofibers that act as efficient heat transfer pathways.



Thermal behaviour

In applications – we need to define service temperature
Could temperature have a critical effect on the material in service?

- Do the properties of the material change with temperature?
- Are the changes reversible?
- Do the changes compromise the safety of the application?

Engineering practice –

Define maximum operating temperature

Define minimum operating temperature

Define a thermal cycle

(define storage conditions and shelf life)

Heat capacity

The ability of a material to absorb heat

Quantitatively: The energy required to produce a unit rise in temperature for one mole of a material by 1K.

heat capacity (J/mol-K) $\rightarrow$

$$C = \frac{dQ}{dT}$$

$\leftarrow$ energy input (J/mol)

$\leftarrow$ temperature change (K)

- Two ways to measure heat capacity:
 - C_p : Heat capacity at constant pressure.
 - C_v : Heat capacity at constant volume.

$$C_p \text{ usually } > C_v$$

- Heat capacity has **units** of J / mol. K

Specific heat capacity



Heat capacity by unit of mass

Specific heat capacity
(J/kg-K) $\rightarrow$

$$c = \frac{dQ}{mdT}$$

energy input
(J/mol) $\leftarrow$

temperature change
(K) $\leftarrow$

Mass (kg) $\nwarrow$

- Specific Heat capacity has units of J / kg.K

Thermal behaviour

- **Atomic Vibrations** due to thermal energy

In solids the vibrations organise into waves or **phonons**

A **phonon** is a quantum of vibrational energy

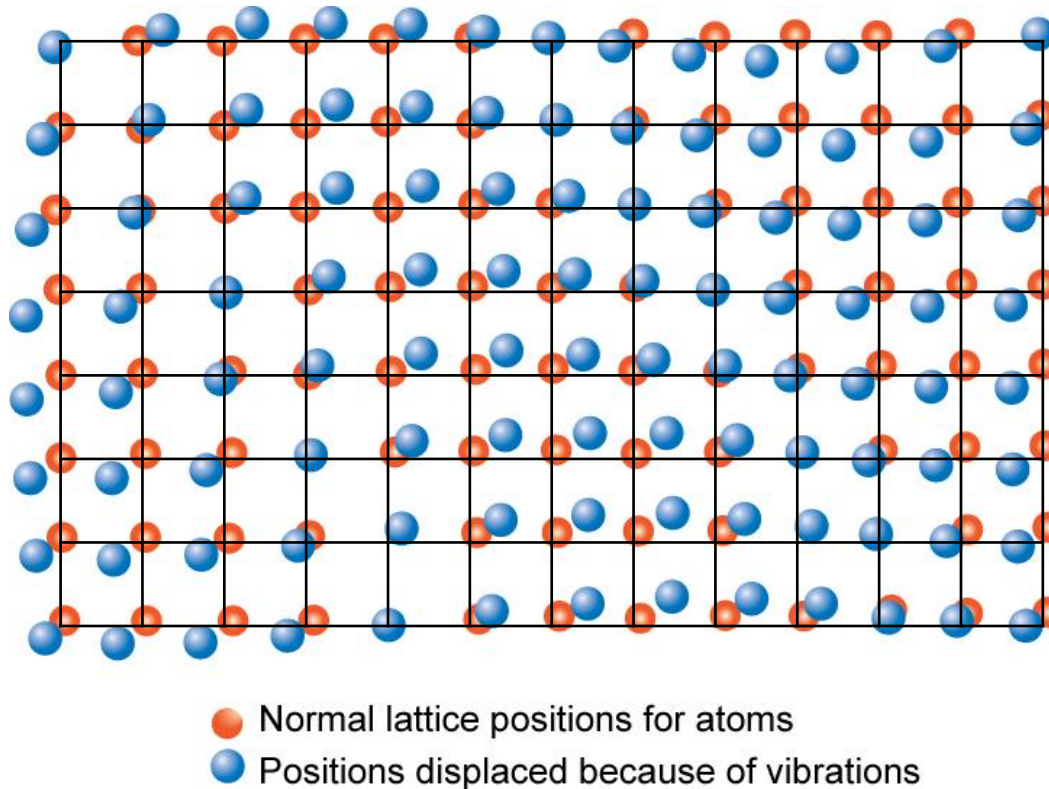


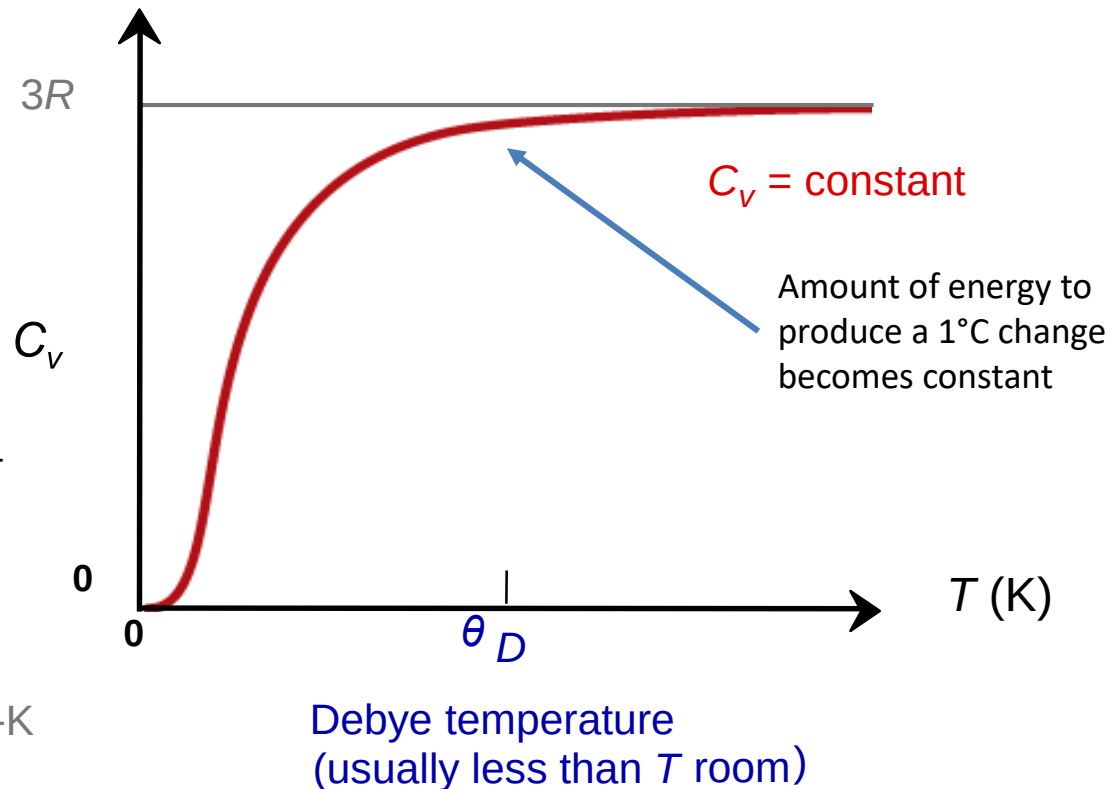
Fig 20.1 Callister & Rethwisch 9e

Temperature dependence of heat capacity

- **At the atomic scale:**

- Energy is stored as atomic vibrations
- As temperature increases, the average energy of atomic vibrations increases

- Heat capacity:
 - increases with temperature
 - for solids it reaches a limiting value of $3R$



R = gas constant = 8.31 J/mol-K

Specific heat

	Material	c_p (J/kg-K) at room T
↑ increasing c_p	• <u>Polymers</u>	
	Polypropylene	1925
	Polyethylene	1850
	Polystyrene	1170
	Teflon	1050
	• <u>Ceramics</u>	
	Magnesia (MgO)	940
	Alumina (Al ₂ O ₃)	775
	Glass	840
	• <u>Metals</u>	
	Aluminum	900
	Steel	486
	Tungsten	138
	Gold	128

c_p (specific heat): (J/kg-K)

C_p (heat capacity): (J/mol-K)

Selected values from Table 20.1,
Callister & Rethwisch 9e.

Thermal conductivity

The ability of a material to transport heat.

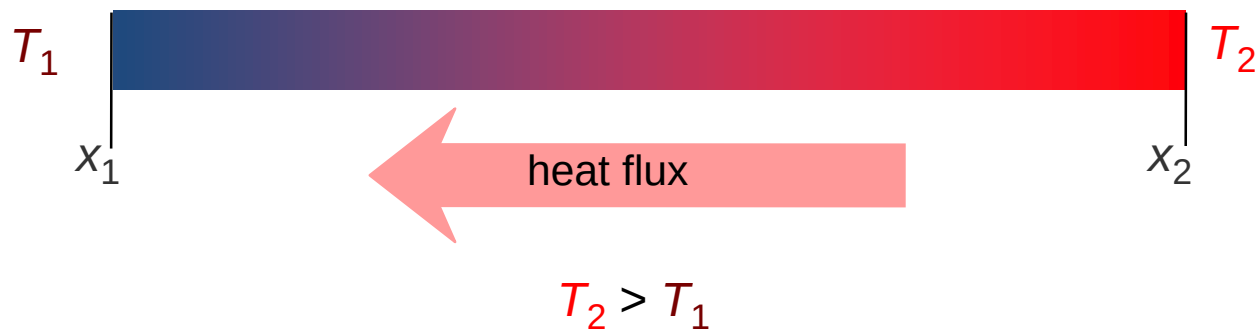
Fourier's Law

$$q = -k \frac{dT}{dx}$$

heat flux (J/m²·s) → q

← $\frac{dT}{dx}$ temperature gradient from hot to cold

← k thermal conductivity (J/m·K·s)



hot to cold = negative temperature gradient

Thermal conductivity

Two mechanisms of heat conduction in solids:

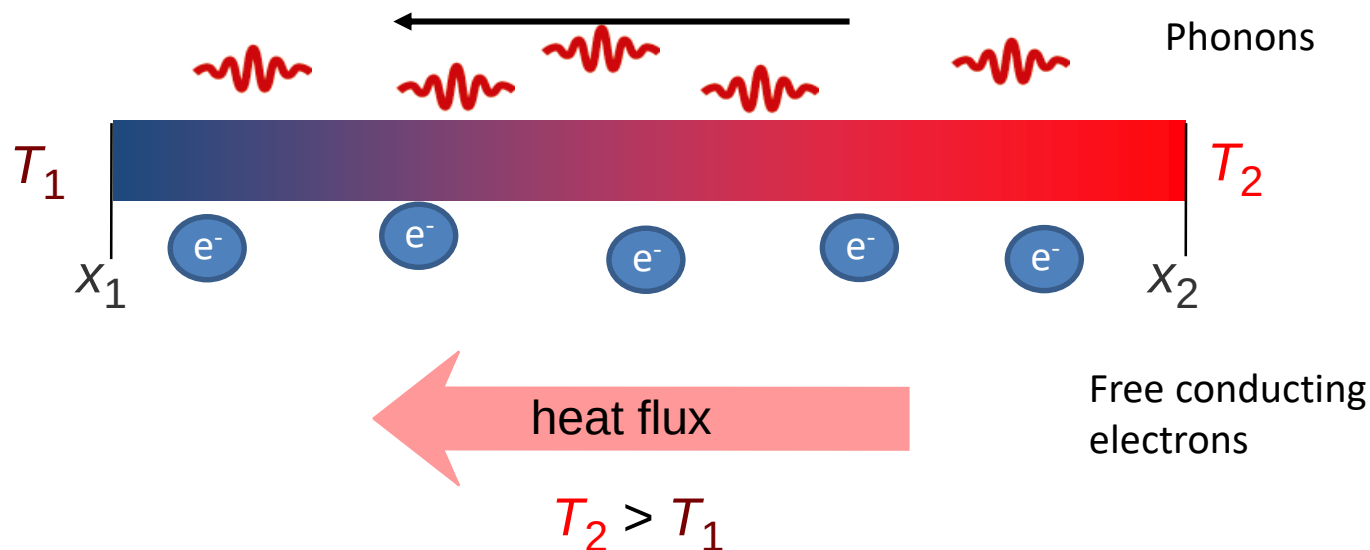
lattice vibration waves (phonons) and

free electrons (in the conduction band)

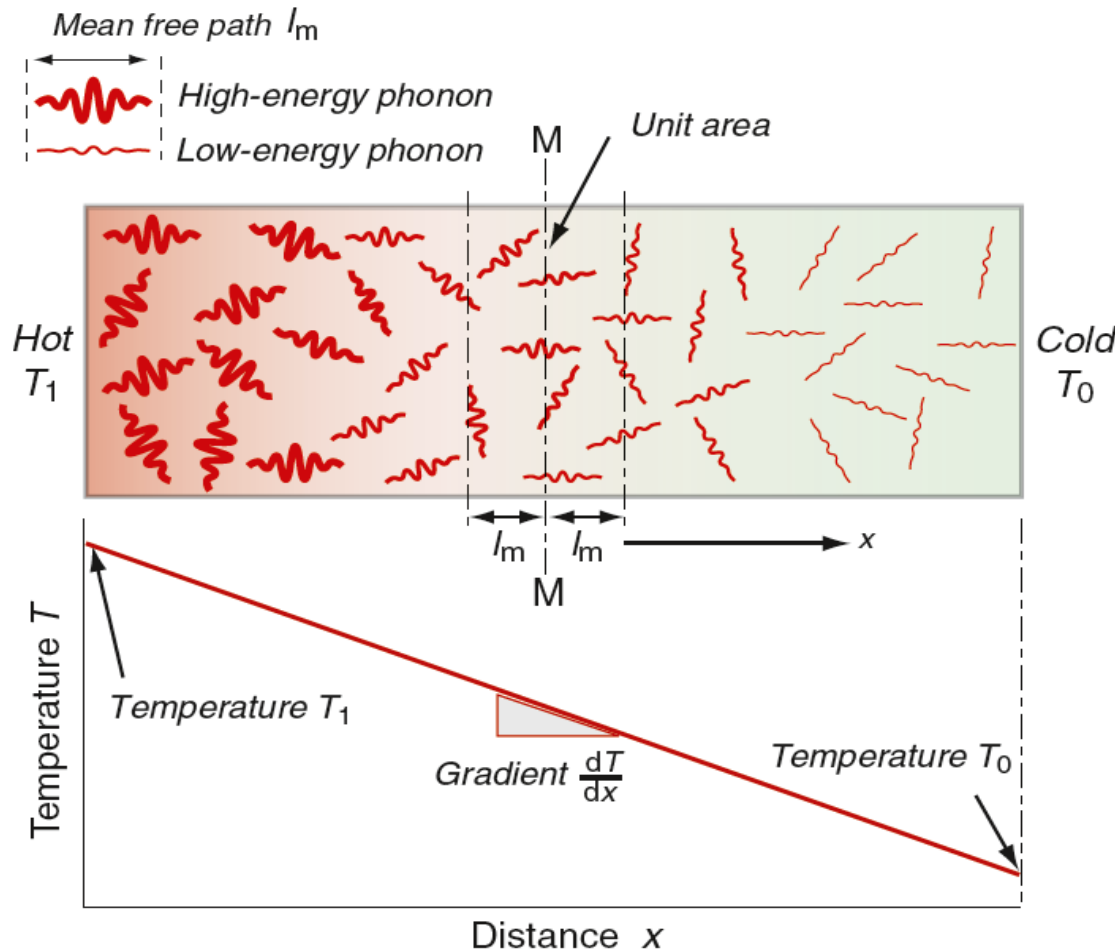
$$k = k_l + k_e$$

Labels for the equation:

- Total thermal conductivity (points to k)
- Lattice thermal vibration (points to k_l)
- Electron thermal vibration (points to k_e)



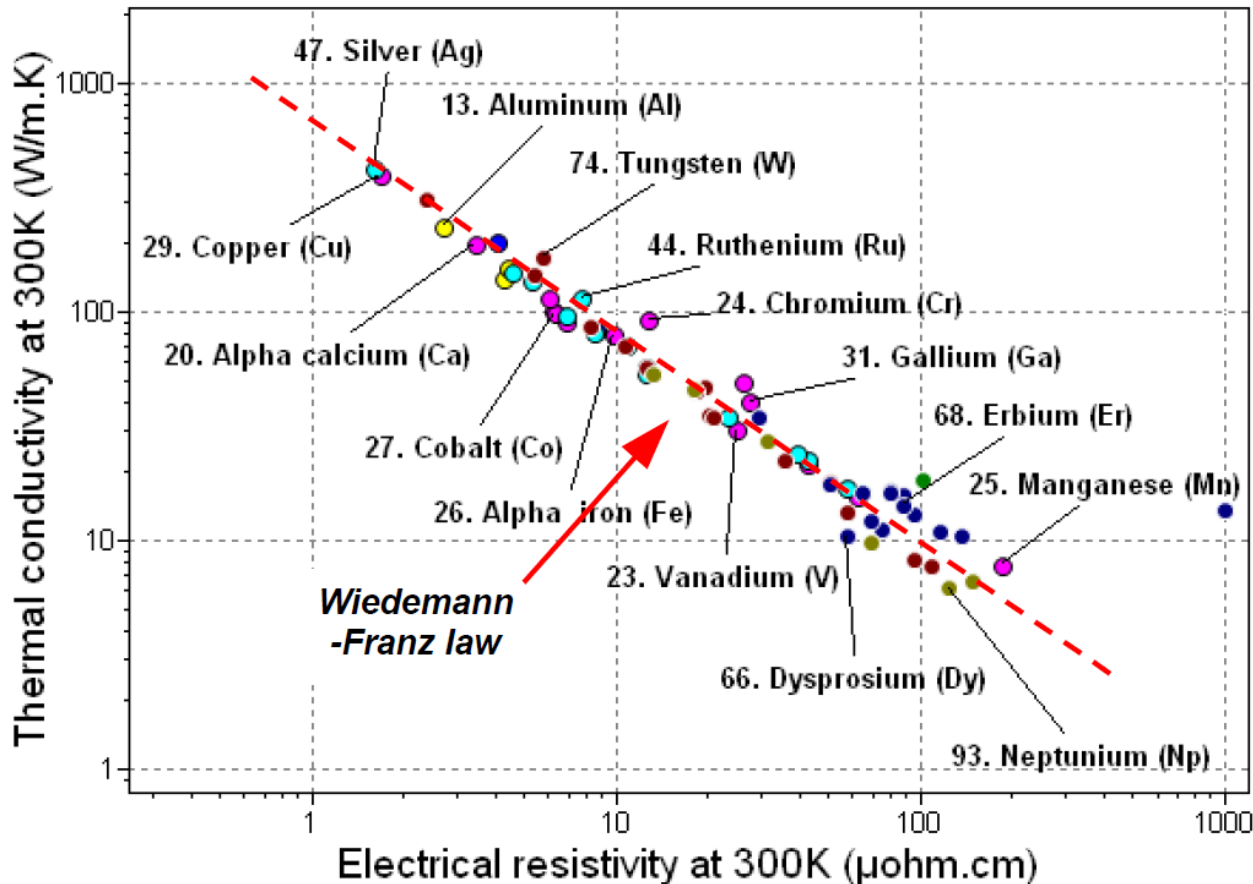
Thermal conductivity



- Atomic vibrations and free electrons in hotter regions transport energy to cooler regions
- When a solid is heated, the heat activates elastic wave packets or phonons
- The phonons travel at the speed of sound but only move a short distance before they are scattered
- The distance a phonon can travel before bouncing off something else is called the mean free path – typically less than $0.01 \mu\text{m}$

Thermal conductivity

In **metals** the dominant heat conductor is the free electrons

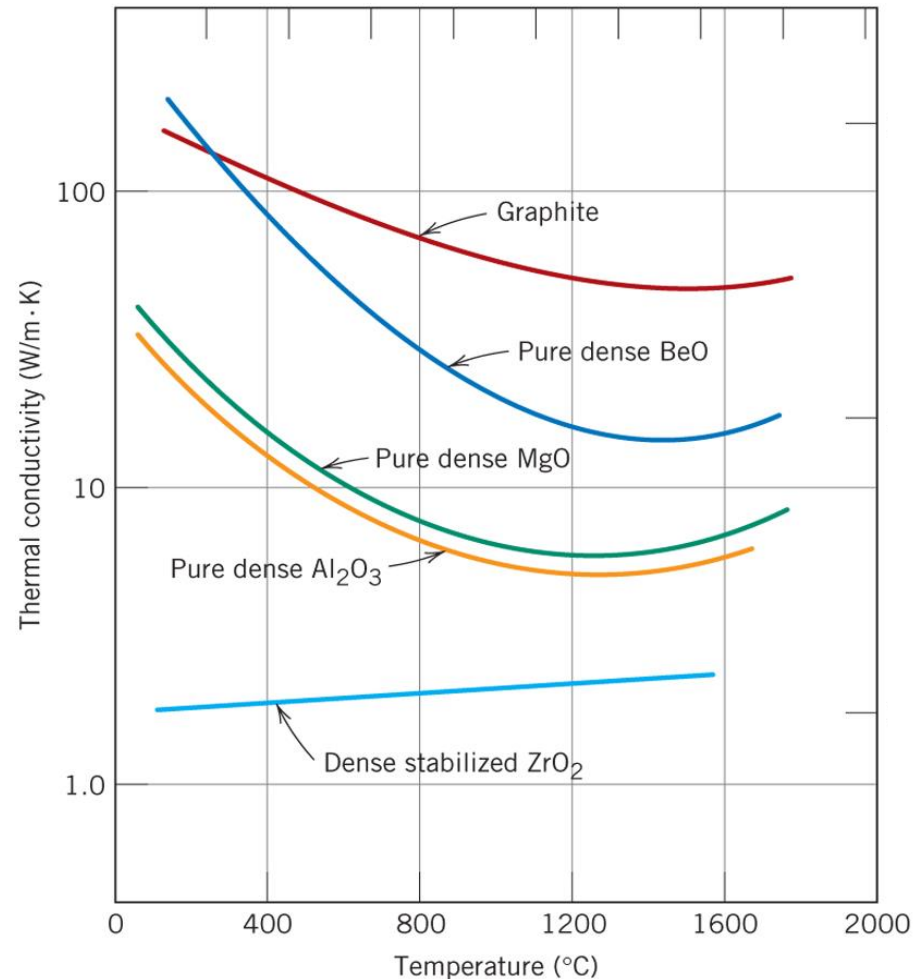


Thermal conductivity and electrical conductivity are ~proportional

Thermal conductivity

In **ceramics** the dominant heat conductor is phonon transport

- k is usually small because there are few free electrons → heat is dominated by phonon transport
- Phonons travel less efficiently → easily scattered
- Amorphous materials have a disordered structure impeding phonons transport
- Increase in temperature causes phonon transport to increase but drops at high temperatures



Thermal conductivity

In **polymers** the dominant heat conductor is molecular vibration



Aerogels

- Very low thermal conductivity
- Energy transfer via vibration/rotation of polymer chains
- Higher crystallinity
- Greater thermal insulation

Thermal conductivity

	Material	k (W/m-K)	Energy Transfer Mechanism
 increasing k	• <u>Metals</u>		
	Aluminum	247	motion of free electrons
	Steel	52	
	Tungsten	178	
	Gold	315	
	• <u>Ceramics</u>		
	Magnesia (MgO)	38	atomic vibrations
	Alumina (Al ₂ O ₃)	39	
	Soda-lime glass	1.7	
	Silica (cryst. SiO ₂)	1.4	
	• <u>Polymers</u>		
	Polypropylene	0.12	vibration/rotation of chain molecules
	Polyethylene	0.46-0.50	
	Polystyrene	0.13	
	Teflon	0.25	

Selected values from Table 20.1, *Callister & Rethwisch 9e*.

Thermal diffusivity

Heat management:

Thermal diffusivity, α , measures the rate of heat transfer in a material from hot to cold

$$\alpha = \frac{k}{\rho c_p}$$

Where

k = thermal conductivity

ρ = density

c_p = specific heat capacity

High values of thermal diffusivity
mean rapid transfer of heat –
good for dispersion of heat

Low values of thermal diffusivity are
good for thermal insulation material

Summary



Thermal properties:

- Heat capacity and specific heat capacity
- Thermal conduction by vibrational (phonon) waves or free electrons (conduction band)
- Conduction by different mechanism in
metals (free electrons)
ceramics (phonons)
polymers (molecular vibration)